

PAPARAZZI UAV / MESH RESEARCH

A Self-Organising Broadcast Mesh for Small UAV Fleets

Paparazzi UAV broadcast mesh research and implementation

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Flighted design, portable architecture: this document describes the current GNSS-assisted TDMA transport with clock holdover and GNSS-independent randomized fallback. The software design targets broadcast-capable mesh radios; the measured reference implementation uses EByte E52-400NW22S hardware. Product-specific numbers are labelled as reference-profile results. Phase 1 investigation of a fully asynchronous coded transport is documented separately in [Asynchronous Coded Mesh for Paparazzi UAV](#). The coded design is currently a simulator and codec experiment; it must not be confused with deployed airborne behavior.

Abstract: One Airspace, Many Tiny Aircraft

Monitoring and controlling one small drone is familiar engineering. Doing the same for a fleet is a different problem wearing the same propellers. Every aircraft must report fresh state, hear occasional operator commands, track its neighbours, survive members joining or disappearing, and do all of that through one narrow half-duplex channel. A naïve design can work beautifully with two aircraft and then collapse at thirteen, which is an awkward time to discover that broadcast packets are not free.

This work presents a self-organising broadcast mesh for Paparazzi UAV. Aircraft learn transmission opportunities from received traffic instead of fixed ID maps, retain timing through bounded GNSS outages, fall back to randomized access when synchronized TDMA is no longer safe, and reduce their own traffic before a radio queue overflows. Compact `MESH_STATE` reports feed traffic awareness and a fail-closed TCAS policy. Ground-side identity recovery, process restarts, and operator link displays are designed as part of the same system rather than as optimistic afterthoughts.

The result is not a claim that one configuration fits every radio. It is a repeatable method: characterize the candidate hardware, model its airtime and forwarding behavior, solve the schedule, simulate churn and clock loss, then verify the complete fleet on a bench before flight. The reference profile was built and tested with EByte E52-400NW22S radios at 434.125 MHz and 62.5 kbit/s. Another radio may be used when it satisfies the interface contract and the full qualification procedure in this guide is repeated with its real parameters.

Keywords: unmanned aerial vehicles; wireless mesh networks; self-organising TDMA; resilient telemetry; low-power lossy networks; GNSS holdover; Age of Information; Paparazzi UAV

Research Questions and Contributions

This work addresses four questions:

1. Can arbitrary aircraft join and leave a shared broadcast channel without a fixed slot-to-identity map or a continuously available coordinator?
2. Can the network preserve bounded, explicitly degraded behavior through short GNSS outages, longer synchronization loss, and process restarts?
3. Can channel load, queue occupancy, and state freshness be made testable properties rather than informal expectations?

4. Can the implementation remain portable across radios without presenting measurements from one product as universal evidence?

The principal contributions are a leased, observation-driven slot allocator; a clock state machine joining GNSS TDMA, monotonic holdover, randomized fallback, and staged recovery; population-aware telemetry and queue admission control; a compact traffic-state path with fail-closed freshness semantics; a restart-tolerant ground identity and health protocol; and an executable qualification ladder separating analytical, simulated, software, hardware, and flight evidence. The work is an engineering design study, not a controlled comparison claiming superiority over every mesh MAC or routing protocol.

Acknowledgements

This work stands on a long chain of scientific ideas: radio propagation, distributed systems, estimation, control, graph theory, collision avoidance, and the patient art of measuring what the hardware actually did. The author thanks the scientists, engineers, reviewers, and experimenters who made those fields usable rather than merely impressive.

Particular thanks go to the developers and maintainers of Paparazzi Autopilot. Their open architecture, generated message system, simulation tools, flight software, and willingness to expose the machinery made this investigation possible. Open-source infrastructure is easy to overlook precisely because it is already there when the difficult question arrives.

In *Eve's Diary*, Mark Twain writes, “It is best to prove things by actual experiment; then you KNOW; whereas if you depend on guessing and supposing and conjecturing, you never get educated.” That is a fair summary of the method used here: model first, test repeatedly, and let the radio have the final word. It usually does, and it has no concern for elegant slides.

Read This First: The System in One Page

The flighted system has two cooperating layers. Paparazzi decides **when an aircraft may originate** a compact `MESH_STATE` packet. The radio network decides **how that packet is forwarded** across the broadcast mesh. The operator does not choose either layer during flight: all peers use one qualified radio profile, while aircraft automatically select `mesh_solo`, `mesh`, or `mesh_manifold` telemetry.

`MESH_STATE` is the primary peer-presence and motion stream. `GPS_LLA` is a slower full GPS report for the GCS. The implementation is divided as follows:

Concern	Source of truth
Airborne MAC, packet encoding, traffic storage	<code>traffic_info.c</code>
Public traffic and TCAS API	<code>traffic_info.h</code>
TCAS freshness and fail-closed policy	<code>tcas.c</code>
GPS, holdover, fallback, and recovery policy	<code>traffic_info_mesh_clock.h</code>
Reference-radio provisioning	<code>e52_provision.py</code>
Capacity and command reserve	<code>mesh_phase_optimizer.py</code>
Join, leave, and collision recovery	<code>mesh_slot_sim.py</code>

Useful background includes the [Paparazzi message documentation](#), and [ITU-R M.1371](#) for the self-organising TDMA ideas behind slot observation and timed leases. For the tested reference hardware, also use the included [EByte E52 user manual](#).

Required radio behavior: transparent PPRZLink transport, broadcast delivery to all reachable peers, multi-hop forwarding or a documented software-relay path, duplicate suppression, bounded queues, and a characterized half-duplex airtime/CSMA model. Equivalent mechanisms with different names are acceptable.

Tested reference profile: EByte E52-400NW22S, channel 24 (434.125 MHz), rate 0 (62.5 kbit/s), 460800 baud UART, 10 dBm EIRP, 0 dBi antennas, all-routing broadcast, and a five-frame transmit cache. These values are evidence for this hardware profile, not defaults for an unknown radio.

Fleet: nine aircraft are common and sixteen aircraft are the normal maximum. Sixty-four aircraft is an addressability and graceful-degradation research target, not a real-time TCAS claim. The ground station is AC_ID 0 and may move on the ground. Aircraft use distinct arbitrary AC_IDs in 1..254; ID 255 remains reserved for broadcast and internal sentinels. No ordering or sequential numbering is assumed.

Priority: fault tolerance and multi-hop coverage first, then the highest state update rate that fits the qualified channel and transmit queue.

Modeled envelope: 13 routing peers use the `mesh` fixed-wing profile at 57.7% modeled channel utilisation. Seventeen peers (16 aircraft and GCS) automatically use the `mesh_manifold` profile at 59.4%. Both use 32 self-organised slots, 25 steady fair-share slots, a 16 s superframe, and a 2 s `MESH_STATE` scheduler ceiling. These are software acceptance baselines, not substitutes for multi-radio bench testing or flight qualification.

1. Two Layers, One Network

1.1 The radio contract

The flighted profile makes every radio, including the GCS, a routing peer. There is no fixed relay aircraft and no single point of failure. A drone may land for a battery swap, disappear, or rejoin without reconfiguring the others. The tested EByte profile selects this behavior with `AT+TYPE=0`; another product may call it flooding, managed broadcast, repeater

mode, or something invented by a marketing department on a Friday afternoon. The name does not matter. The measured forwarding behavior does.

This distinction matters:

- **Paparazzi decides when a node originates a new frame.** `traffic_info.c` uses GNSS-synchronised self-organising TDMA, bounded clock holdover, GNSS-independent randomized fallback, and radio-queue back-pressure.
- **The radio network decides how the frame is forwarded.** The required behavior is bounded all-to-all broadcast without an application-level ACK storm. On the reference hardware, `AT+OPTION=3` makes every routing peer forward a newly seen broadcast once and its duplicate filter stops loops.
- **Broadcast and unicast routing are different services.** A radio may offer RSSI-selected routes for unicast while using duplicate-suppressed flooding for broadcast. Qualification must measure the mode actually used by `MESH_STATE` ; a glossy unicast range claim is not a broadcast timing model.

For position exchange, broadcast is retained because every aircraft must learn about every other aircraft and because it continues working without a coordinator or acknowledgement exchange. Directed, high-volume application traffic may use a candidate radio's qualified unicast service when practical, so one recipient does not incur an all-peer flood.

The two control layers must not be confused:



The TDMA allocator controls only the arrow from `traffic_info` to the local radio. In the reference profile, forwarded copies are created inside the radios and do not require a Paparazzi slot of their own. Their airtime is still charged to the shared channel, because physics has declined every request to support free relays. A software-relay radio can also be used, but its forwarding work, queueing, duplicate suppression, and CPU cost must be modeled explicitly.

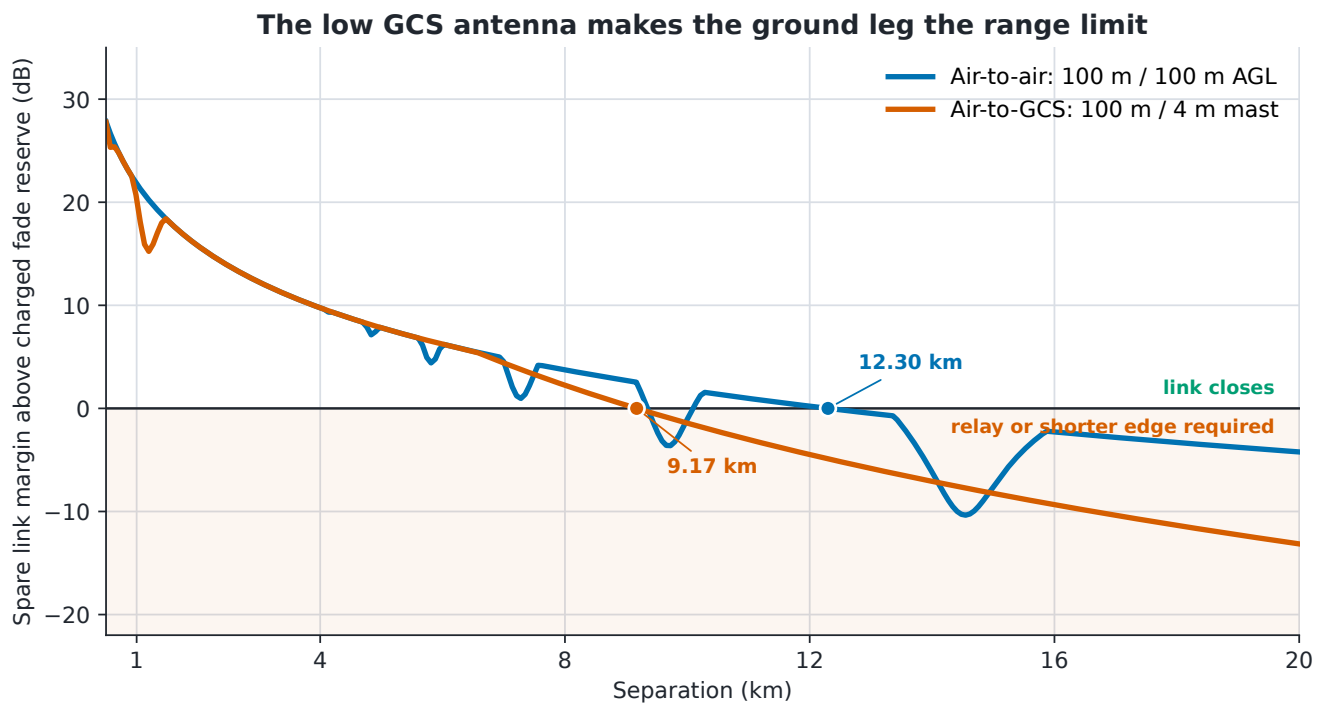
1.2 Why every reachable peer may need to help

The original 3 x 3 km area is direct-link friendly: its 4.24 km diagonal closes with at least 7.3 dB spare margin above the already charged 10 dB fade margin. The future operating areas are not:

Area	Diagonal	Worst direct margin
3 x 3 km	4.24 km	+7.3 dB
11 x 11 km	15.56 km	-14.7 dB
17 x 3 km	17.26 km	-16.9 dB

These values come from `mesh_link_budget.py` using the exact two-ray model, 434.125 MHz, a 4 m GCS mast, 50-120 m aircraft altitude, 45 degree bank, 10 dB fade margin, and the hard 10 dBm EIRP limit. The expanded areas need short air-to-air hops through the swarm; a terminal-only aircraft at an edge can partition the topology.

The deterministic design ranges from the same model are 12.30 km for an air-to-air link at 100 m AGL and 9.17 km for an aircraft-to-GCS link with the 4 m mast. These are not universal radio ranges: they include the stated antenna, bank, feedline, terrain-clearance, and fade assumptions. Mission spacing must use the weaker link applicable to each edge and retain positive margin; “all nodes route” does not make an out-of-range edge exist.



Analytical model: 434.125 MHz, 10 dBm EIRP, 45 deg bank, 10 dB fade reserve, average grass-covered ground.

Figure 1. Analytical spare margin after the stated 10 dB fade reserve has already been charged. The zero crossing is a model boundary, not a measured radio range; terrain, interference, installation, and regulation still require site-specific qualification.

1.3 Broadcast is robust, not free

All-router broadcast is expensive. One 30-byte `MESH_STATE` frame occupies 7.50 ms per RF transmission. The conservative capacity model charges one transmission at every one of the 13 routing peers for each originated broadcast:

An individual flood can use fewer transmissions when some peers are unreachable or receive only duplicates, but capacity is not budgeted on that optimistic case. The 13-transmission “flood tax” is a planning upper bound for the connected all-router fleet, not a claim that every packet always produces exactly 13 observable RF transmissions.

The tested radio has only five transmit-cache entries. On overflow it clears the whole cache, creating a network-wide telemetry gap. The reference schedule is therefore designed around measured flood cost and queue behavior, not around the nominal 62.5 kbit/s PHY rate. A deeper queue is not automatically faster; it may simply store a longer delay with greater dignity.

`mesh_phase_optimizer.py` proves the delivered profile:

Check	Result
Nodes / routing nodes	13 / 13
Flood tax	13 transmissions
Modeled channel utilisation	57.7% (operator ceiling 60%)
Flood span, mean + 3 sigma	290.0 ms
Flood span, absolute modeled maximum	357.6 ms
Physical slot / guarded usable gap	500.0 / 478.0 ms (+34%)
Peak reference-radio cache depth	1 of 5
Cache overflow events	0

The reference-radio documentation specifies no safe aggregate channel-utilisation limit, and no hardware measurements establish one for this network. Utilisation is therefore reported rather than compared with a default pass/fail percentage. The optimizer accepts an explicit `--utilisation` ceiling when an operator has a measured limit for a particular deployment. The actual software acceptance checks are that a flood fits its slot, generated messages do not burst together, the modeled five-frame cache does not overflow, and slot collisions remain bounded in the churn simulation. Applicable national limits on frequency, bandwidth, power, and duty cycle remain separate and must be checked for the flight location.

The optimizer's deliberately unscheduled control experiment reaches 17 cached frames and overflows. Phase placement and TDMA are safety mechanisms, not cosmetic tuning.

The former 375 ms slot was only the arithmetic result of dividing a 12 s superframe into 32 physical slots; it was not derived independently from radio flood latency. For the common 13-node reference model, one 30-byte `MESH_STATE` transmission is 7.50 ms and the modeled flood spans are 227.6 ms mean, 290.0 ms mean plus three standard deviations, and 357.6 ms absolute maximum.

The dense 17-node model gives 369.0 ms mean plus three standard deviations and 467.6 ms absolute maximum. The delivered 500 ms slot reserves 10 ms for the 100 Hz gate and 12 ms for clock skew, leaving 478 ms and a 10.4 ms margin over that absolute modeled bound. Both calculations still assume that every relay adds one independent uniform 0..20 ms CSMA delay and that the inferred 13-byte reference-radio mesh header is correct. Hardware flood-latency measurement is required before treating the result as a physical guarantee.

2. A Fleet That Organises Itself

2.1 Joining without a slot map

The superframe is 16 s with 32 slots of 500 ms. Slots are not derived from `AC_ID`. A joining peer:

1. listens for three complete superframes;
2. learns slot occupancy from ordinary `MESH_STATE` arrival times;
3. claims a demonstrably free slot;
4. yields deterministically if a lower `AC_ID` is already using that slot;
5. leases secondary and primary slots so mutually deaf collisions eventually separate even when nobody can decode the collision.

`MESH_SLOT_FREE` is `0xFF`, not zero. `AC_ID 0` is the GCS and is a legitimate mesh slot owner, while aircraft IDs are limited to 1..254. The identity domain therefore addresses at most 255 members (GCS plus 254 aircraft), so byte-sized member counts remain sufficient without changing the over-air format. The delivered 32-slot profile supports at most 32 simultaneous full-membership peers. The GCS participates in routing and traffic visibility but is deliberately excluded from TCAS collision avoidance; a future stationary-obstacle policy could use its track explicitly.

Thirty-two slots are intentional. Thirteen peers could fit in sixteen, but the churn test showed that sixteen slots were too tight when up to six aircraft leave and rejoin. With 32 slots, eight deterministic 4000-frame simulations at 5% churn all pass; collision occupancy is 2.18-3.75%, every collision is transient, and no node loses its primary slot.

There is no reservation packet. Each receiver infers the sender's slot from GNSS-aligned or bounded-holdover arrival time. Listen-before-claim avoids cold-start over-allocation. Round-robin expansion prevents two nodes from grabbing the same apparently free slot in one superframe. Timed leases break collisions that are otherwise invisible because collided frames cannot be decoded.

When GNSS time disappears, the node retains the last GNSS-to-monotonic anchor for up to 60 seconds instead of jumping to boot-relative time. The primary reservation and absolute superframe phase therefore survive a short outage; position-invalid policy contracts opportunistic slots. This requires no RTC: the autopilot's monotonic clock carries the startup GNSS epoch while power remains applied. The bound assumes at most 100 ppm error per node: two worst-case clocks separate by about 12 ms in one minute. That skew and the 50 ms scheduler sampling interval fit inside the common 13-node statistical budget, but not the dense 17-node budget.

One superframe before that bound, the denied node and peers which hear its `HOLDOVER` advertisements clear slot authority and enter `MESH_CLOCK_ASYNC`. This fleet-wide downgrade prevents synchronized and asynchronous MACs from competing indefinitely during partial GNSS loss. Each peer then originates from its monotonic local clock with randomized CSMA timing. A trusted pre-outage membership view selects 4-8 seconds for a solo aircraft, 8-16 seconds for two to four aircraft, and 16-24 seconds for five or more aircraft. A node that rebooted without synchronized membership always uses the conservative 16-24 second range. Learned population never decreases during an outage because packet loss is not proof that a peer departed. Asynchronous frames remain valid presence traffic but never reserve a TDMA slot. GNSS-ready fallback peers advertise `RECOVERY` without extending the denied-peer lease. `RECOVERY` carries a deterministic absolute GPS-frame target: every peer selects the same next eight-frame epoch at least seven frames ahead and adopts any later target it receives. This lets rebooted and late-joining peers converge before TDMA resumes. At that target, two seconds of continuous valid GPS and the normal listen-before-claim cycle are required before transmission. Coherent recovery during early holdover preserves slots, while a correction over 10 ms clears them.

2.2 Spend airtime where it matters

All radios route continuously at the network layer. Paparazzi adapts only the rate at which a node originates its own state:

- a healthy airborne node with GNSS or bounded holdover and valid position receives its fair share of available slots;

- a landed or position-invalid node keeps one heartbeat slot;
- a node without bounded network time uses population-aware randomized access instead of pretending that boot-relative slots are synchronized;
- HOME, emergency, failsafe, and low-battery states request one extra slot;
- when the estimated radio queue reaches the high-water mark, the node gives up one opportunistic slot before the qualified hardware can overflow;
- when aircraft leave, healthy peers gradually claim quiet slots; when they return, peers contract immediately.

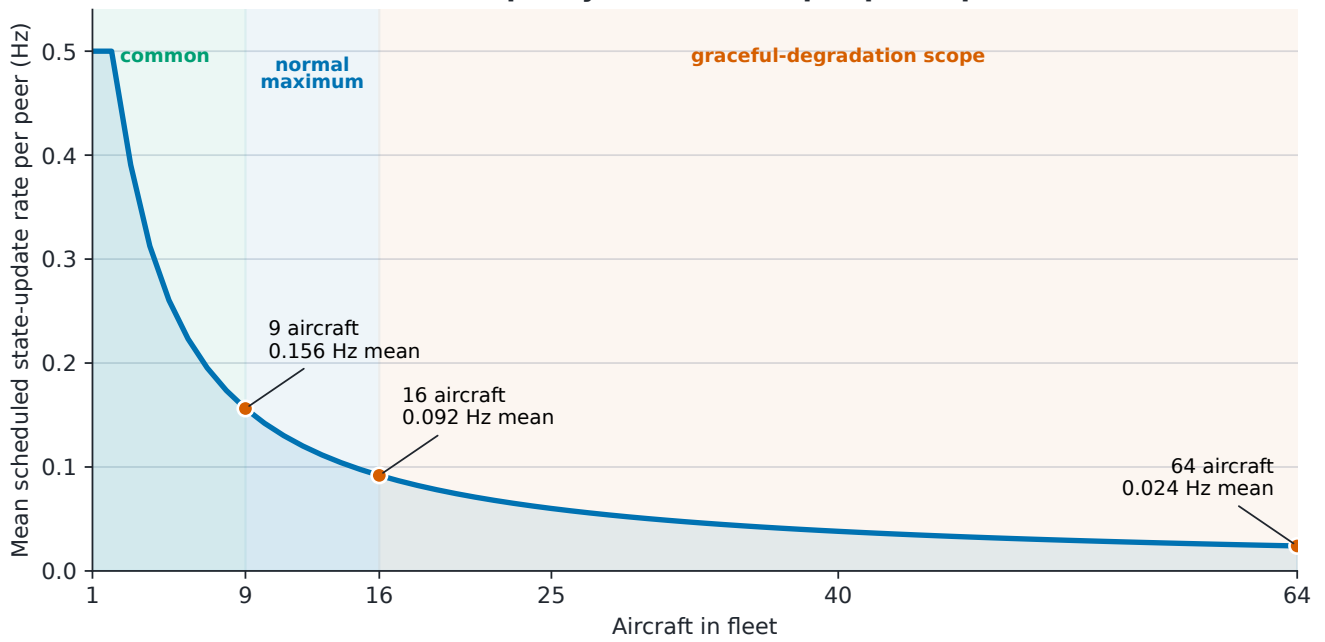
Of the 32 physical slots, 25 are distributed as steady-state fair share and seven remain as headroom for joins, partitions merging, and temporarily different membership views. For fair-share slots and live nodes, each healthy node receives a baseline of slots. The first sorted AC_ID ranks receive one remainder slot, so quotas differ by at most one and sum to all available slots. The winner window advances by one rank every 81 superframes, about 21.6 minutes, giving long-term fairness without fleet-wide claim churn.

At 13 peers, twelve are entitled to two slots and one to one slot per 16 s. The average entitlement is therefore Hz instead of the floor-only 0.0625 Hz. At the nominal nine-peer population, seven peers receive three slots and two receive two, for an average entitlement of 0.174 Hz. Collision-healing leases and policy contraction can make observed rates lower than these steady-state entitlements. Sparse fleets may now use up to `MESH_TDMA_MAX_REUSE=8` slots. The cap does not add slots or shorten them: it lets fewer live peers use slots that would otherwise remain empty. Theoretical steady-state entitlements are shown below; counts include the GCS because it is a real mesh peer:

Live peers	Reuse 4	Reuse 8	Improvement
1	0.25 Hz	0.50 Hz	100%
2	0.25 Hz	0.50 Hz	100%
3	0.25 Hz	0.50 Hz	100%
4	0.25 Hz	0.39 Hz	56%
9	0.17 Hz	0.17 Hz	unchanged
13	0.12 Hz	0.12 Hz	unchanged

Collision-healing leases deliberately limit dense-fleet convergence, so the measured rates are lower than the mathematical entitlement. This revision does not weaken those leases merely to report a higher number. Emergency state still receives priority within the same bounded channel budget.

Fixed channel capacity forces lower per-peer update rates



Model: 25 fair slots per 16 s superframe; GCS included as one peer; maximum 8 slots per peer.

Figure 2. Theoretical steady-state fair-share rate from 25 slots per 16 s superframe, with the GCS counted as a peer and reuse capped at eight. Lease healing, synchronization loss, invalid position, and queue pressure can reduce observed rates below this curve.

The 2 s telemetry period is an enabling ceiling, not a promise that every node transmits every 2 s. Actual emission remains controlled by live slot ownership, position validity, synchronization state, and cache pressure.

2.3 The constants that keep everyone honest

Parameter	Value	Reason
MESH_TDMA_SUPERFRAME_MS	16000	32 absolute-model flood-safe slots
MESH_TDMA_NB_SLOTS	32	13 peers plus churn headroom
MESH_TDMA_FAIR_SLOTS	25	steady quota; seven slots absorb membership disagreement
MESH_TDMA_MAX_REUSE	8	bounded sparse-fleet acceleration
MESH_ENTRY_FRAMES	3	listen before transmitting
Secondary lease	6-13 frames	break secondary collisions
Primary lease	120-239 frames	avoid healthy-fleet churn while still breaking silent primary collisions
Slot age	4 frames	tolerate loss, reclaim departed nodes
Remainder epoch	81 frames	rotate one rank after ageing and serialized expansion settle
Holdover	60 s	preserve GPS epoch while worst-case relative drift stays bounded
GNSS acquisition	2 s	reject fix flapping before entering TDMA
Position holdover	16 s	allow one superframe of globalized estimator output, then fail closed
Asynchronous interval	4-8 / 8-16 / 16-24 s	solo / 2-4 aircraft / larger or unknown fleet
Recovery epoch	8 frames	deterministic common target beyond all fallback leases
Cache high water	3 of 5	local-only admission guard; reserve two entries for hidden relay/telemetry traffic

The static assert in `traffic_info.c` forces the generated `MESH_STATE` period to equal `superframe / max reuse` ($16 / 8 = 2$ s) in either supported telemetry mode layout. A mismatched telemetry file therefore fails the aircraft build instead of failing in flight.

The cap and telemetry period are a fleet-wide protocol profile. Update every participating aircraft before using the faster profile; do not mix reuse-four and reuse-eight firmware in one mesh.

2.4 One profile, three automatic operating modes

There is no operator mode selection, custom wire protocol, additional ground application, or in-flight radio reconfiguration. Every scenario uses the same qualified routing/broadcast profile. Aircraft start in `mesh`, then select one of three generated telemetry modes from observed peers:

Mode	Automatic condition	Purpose
mesh_solo	GCS contact, no aircraft peer, 16 s quiet	5 Hz direct-link state for the common ≤ 1 km fallback
mesh	up to 11 observed aircraft peers	normal 1-12 aircraft operation
mesh_manifold	enter at 12 peers, leave at 10	bounded 13-16 aircraft operation with hysteresis

Telemetry starts in `mesh` so useful flight state reaches the GCS promptly. The independent TDMA network-entry logic still listens for three superframes before claiming slots, preventing newly powered peers from making conflicting slot assumptions before their membership maps converge.

The OCaml link sends one targeted `PING` every five seconds to the live aircraft that was least recently probed. The addressed aircraft returns one `PONG`. Identity recovery is deliberately separate: an unregistered live aircraft is sent a serialized `ALIVE_REQ` and returns one MD5-bearing `ALIVE`. This avoids the adjacent `ALIVE` / `PONG` response burst that previously made normal health probes less reliable. Identity requests use exponential backoff and stop when the server's authoritative `AIRCRAFTS` snapshot contains the aircraft. `NEW_AIRCRAFT` is only a low-latency hint; periodic snapshots make link-first startup, server-first startup, missed lvy events, and either process restarting converge to the same state. Incoming `MESH_STATE` also refreshes the ground link's live-aircraft table without being republished on lvy. Because broadcast traffic is heard by every reachable mesh member, each aircraft can answer two questions locally:

- has the GCS recently pinged me;
- has the GCS recently pinged another aircraft.

The airborne selector chooses `mesh_solo` when the GCS has recently pinged this aircraft, no recent `PING` targeted another aircraft, no peer owns a live mesh slot, and no peer frame has been received for 16 seconds. A synchronized mesh clock is deliberately not required for this one-aircraft case: there is no peer TDMA schedule to coordinate, and requiring GPS time would leave an indoor or GPS-denied bench test permanently on sparse telemetry. The `MESH_STATE` canary continues using the normal GPS, bounded-holdover, or randomized asynchronous mesh timing. A received peer `ALIVE` or `MESH_STATE` leaves solo mode immediately for `mesh`; peer-count hysteresis independently selects `mesh_manifold` only at the high-population threshold. `ALIVE` is weak presence evidence only: it never creates TCAS kinematics, assigns a slot, or shortens the three-superframe listen-before-claim interval. A manual selection of an unrelated diagnostic mode is preserved and disables automatic switching until a mesh mode is selected again.

The ground link probes only aircraft that are still live. Without that rule, an aircraft which landed hours earlier would remain in the link table and suppress solo mode forever. State from an unregistered live aircraft triggers an immediate serialized `ALIVE_REQ`; if that request or response is lost, retries start at four seconds and back off to at most 30 seconds. The four-second retry floor keeps the modeled 17-node recovery envelope below the 60% operator channel ceiling. This closes the historical deadlock where the server waited for an `ALIVE` that could have been lost while the radio booted. Stale aircraft are not eligible for discovery. `ALIVE` remains a discovery and configuration-identity message, not the high-rate heartbeat. `MESH_STATE` is the authoritative aircraft presence and motion stream.

Two `ALIVE` paths coexist intentionally. A phase-spread boot/30-second airborne retry gives passive startup discovery, while a targeted `ALIVE_REQ` gives the ground link deterministic recovery when that unsolicited packet was missed. Both produce the same MD5-bearing `ALIVE`; neither changes the health-probe contract that one `PING` produces only one `PONG`.

The only added wire message is the empty `ALIVE_REQ` datalink request (ID 195), chosen outside IDs used by the repository's older/custom datalink schemas. `Makefile.ac` makes aircraft generation depend on the generated PPRZLink protocol header, so changing `conf/messages.xml` regenerates protocol headers before computing the aircraft configuration MD5. This prevents firmware from advertising a stale identity after message-schema changes. Fixed-wing sends `MINIMAL_COM` at 5 Hz, `ATTITUDE` at 2 Hz, `ENERGY` at 1 Hz, `DATALINK_REPORT` at 0.5 Hz, and `ALIVE` at 0.2 Hz. Rotorcraft uses the same schedule with native `ROTORCRAFT_FP` instead of `MINIMAL_COM`. `MESH_STATE` remains active as the safety/discovery canary in both modes. The mesh transport becomes ready on the first telemetry scheduler tick. Startup then sends one standard `ALIVE` within an AC_ID-derived 250 ms window, waits at least one 60 ms reference-radio drain interval, and sends one registered primary-state callback within a second 160 ms window. Thus the common single join submits useful GCS state in under 0.5 seconds without duplicating either firmware's state serializer. One

independently spread state recovery follows within four seconds for simultaneous fleet power-up, where the first burst may collide. `ALIVE` is retried every 30 seconds until the first GCS PING, allowing the normal server/link discovery cycle to start without a custom handshake.

The primary period was reduced from 250 ms to 200 ms, giving exactly 25% more scheduled state updates and 20% less maximum update latency. The auxiliary rates remain deliberately unchanged: the available solo-link margin is spent on the position and motion data that benefits from freshness, not duplicated diagnostic traffic.

With two routing radios, the repository PHY model places the complete 5 Hz fixed-wing profile at 15.0% aggregate channel occupancy and 7.5% transmit duty per radio. The larger rotorcraft state frame gives 17.9% aggregate occupancy and 8.9% duty. Compared with 4 Hz, the extra state frame per second costs only 1.82 percentage points of aggregate occupancy for fixed-wing and 2.40 points for rotorcraft.

Live NPS measurements changed from 4.158 Hz to 5.006 Hz for fixed-wing `MINIMAL_COM`, a 20.4% observed increase, and from 3.976 Hz to 5.008 Hz for rotorcraft `ROTORCRAFT_FP`, a 26.0% increase. Mean delivered intervals were 199.76 ms and 199.68 ms respectively. In a simultaneous 5 Hz-capable fixed-wing/rotorcraft test, both aircraft appeared in the ordinary link table and a fully settled ten-second window contained zero solo-rate frames.

The design deliberately accepts less throughput than a custom compact packet. In return it uses standard Paparazzi messages, standard generated telemetry modes, the existing PING/PONG path, one small firmware-neutral selector, and no new process for the operator to start.

2.5 A ground link that tells the truth

The serial layer supports 460800 baud and requests exclusive ownership with `TIOCEXCL` where the platform provides it. This matters because two link processes reading one radio split the byte stream and look like RF or parser loss. A second opener now fails clearly instead. Transient `EAGAIN` is ignored; terminal descriptor errors retire the old read watch, close the descriptor, probe for up to 30 seconds, and re-exec the link after the device returns so no stale parser or event-loop state survives a USB reconnect.

PING round-trip time uses `CLOCK_MONOTONIC` and is published only after a matching PONG completes a measured normal probe. Discovery requests do not overwrite RTT state, and delayed or unsolicited PONGs cannot create a negative latency by being subtracted from a newer wall-clock PING.

`DATALINK_REPORT.uplink_lost_time` retains its historical wire name, but the value is **not** a boolean loss alarm. It is the aircraft's whole-second age since the last complete accepted primary uplink frame. The counter resets before command-specific parsing, so a valid PING or waypoint command proves reception even if a command later has no effect. The server forwards the value unchanged. The cockpit therefore labels it `Uplink age [s]` and explains that a nonzero value is normal. In `mesh_solo`, a healthy link commonly cycles around 0-5 seconds because the normal PING cadence is five seconds. A real loss alarm must be a separate state with an explicitly chosen operational threshold.

2.6 Commands first, diagnostics later

The priority order is:

1. operator control and its acknowledgement: `BLOCK -> NAVIGATION`, `MOVE_WP -> WP_MOVED`, and `SETTING -> DL_VALUE`;
2. `MESH_STATE`, which supplies peer presence and compact kinematics to `traffic_info` and TCAS;
3. `GPS_LLA`, the slower authoritative GPS diagnostic for the GCS;
4. `ALIVE` and periodic navigation/health recovery snapshots;
5. optional diagnostics.

Control messages are event-driven and are never assigned a telemetry `phase`. The ground link writes forwarded commands immediately, and the fixed-wing parsers send their standard acknowledgement immediately after applying the change. Periodic copies remain only for eventual recovery. The optimizer reserves one worst-case command/acknowledgement pair every 20 seconds across the fleet plus one adaptive `PING/PONG` pair every five seconds. With solved phases the modeled radio queue peaks at one frame, so an operator command can sit behind at most the frame already being transmitted; future periodic traffic does not form a queue in front of it.

`phase` remains valuable for periodic traffic: generated offsets prevent a single aircraft from presenting a multi-frame UART burst to its five-entry radio queue. It cannot schedule or accelerate asynchronous commands.

3. Qualifying the Radio Layer

The mesh algorithm is portable; a radio profile is not. The flight-tested profile below is for the **EByte E52-400NW22S**. It is included so the deployed system is reproducible and so a replacement has a precise baseline to beat. These commands are not a generic radio API:

```
AT+PANID=250,1
AT+SRC_ADDR=1000+AC_ID,1
AT+TYPE=0           all nodes route, including AC_ID 0
AT+RATE=0           62.5 kbit/s only
AT+CHANNEL=24,1     434.125 MHz
AT+POWER=10,1       10 dBm conducted with 0 dBi antenna
AT+OPTION=3,1       broadcast
AT+DST_ADDR=65535,1
AT+ROUTER_SAVE=0    do not persist a moving topology
AT+ROUTER_CLR=1
AT+ROUTER_SCORE=3
AT+HEAD=0           PPRZLink already carries framing and sender ID
AT+BACK=0           do not inject radio status text into PPRZLink
AT+CSMA_RNG=20      manual minimum; TDMA separates originations
AT+RESET_TIME=0     no periodic RF reset in flight
AT+FILTER_TIME=3000 duplicate suppression
AT+UART=460800,8N1
AT+RESET
```

`AT+TYPE` must follow `AT+SRC_ADDR` because it rewrites the address type bit. The provisioning tool enforces this ordering, checks every response, reopens the serial port after changing baud, and reads the configuration back.

Program an aircraft or the GCS:

```
python3 sw/tools/mesh/e52_provision.py \
  --port /dev/ttyUSB0 --ac-id 129 --factory-reset

python3 sw/tools/mesh/e52_provision.py \
  --port /dev/ttyUSB0 --ac-id 0 --factory-reset
```

Both commands default to `AT+TYPE=0`. A constrained terminal/router experiment is still possible with `--relay-ids 125`, but it is not the delivered flight architecture.

The power rule is absolute: **10 dBm EIRP maximum**. The NW22S PA can produce 22 dBm, but that setting must never be used here. `e52_provision.py` derives conducted power from the EIRP ceiling minus antenna gain and refuses an illegal combination.

3.1 The capability contract for another radio

A candidate radio does not need EByte commands, LoRa modulation, or the same queue depth. It does need an equivalent, measured service:

- transparent transport of complete PPRZLink frames without rewriting sender identity or payload;
- broadcast reception by every reachable peer, including AC_ID 0 at the GCS;
- bounded multi-hop forwarding, either inside the radio or explicitly in Paparazzi, with duplicate suppression and a known hop limit;
- half-duplex and carrier-access behavior whose worst and statistical delays can be measured;
- a documented maximum frame size, UART rate, queue depth, overflow policy, and reconnect behavior;
- legal frequency, bandwidth, power, duty cycle, and antenna configuration for the actual operating region;
- identical fleet-wide settings, or a compatibility mechanism proven under mixed versions.

“It has mesh in the product name” is not an acceptance test. Neither is a point-to-point throughput screenshot. This design depends on all-to-all broadcast timing under simultaneous routing load.

3.2 Porting the profile without guessing

Use this sequence when introducing another radio:

1. **Write down the forwarding model.** State whether broadcast is flooded, selectively relayed, source-routed, or repeated by Paparazzi. Identify what creates and suppresses duplicates and whether forwarded frames are visible to flight software.
2. **Measure one frame.** For every relevant PPRZLink length, record UART-to-air latency, RF time-on-air, CSMA/backoff distribution, and receive-to-forward delay. Repeat near sensitivity and under contention; averages alone are charming but insufficient.
3. **Measure the queue.** Determine usable depth, drain time, overflow policy, and whether hidden relay traffic shares the same queue as local traffic. Update `MESH_MODEM_DRAIN_MS` and `MESH_CACHE_HIGH_WATER` only from these data.
4. **Update the analytical model.** Teach `mesh_phase_optimizer.py` the new PHY airtime, framing overhead, flood tax, forwarding-delay distribution, queue depth, and operator utilisation ceiling. If the assumptions cannot be represented, extend the model before generating a schedule.
5. **Re-solve, do not copy.** Run the phase optimizer for every supported fleet size and telemetry mode. A schedule proven for a 62.5 kbit/s, five-entry reference queue says nothing reliable about a different radio.
6. **Repeat network simulations.** Run `mesh_slot_sim.py` because timing changes can alter collision recovery; run `mesh_gps_denied_sim.py` for fallback load; update and run `mesh_link_budget.py` and `mesh_link_sim.py` with the new frequency, sensitivity, power, antenna, propagation, and relay model.
7. **Create provisioning and readback.** Add a product-specific tool or adapter that applies one fleet profile, verifies it, and fails loudly on mismatches. Do not stretch `e52_provision.py` into pretending unrelated commands are the same abstraction.
8. **Build and bench the whole fleet.** Repeat the acceptance procedure in Section 6 with instrumented packet traces, joins, removals, commands, GNSS loss, queue stress, USB interruption, and the maximum supported peer count.

9. **Qualify in controlled flight.** Start below the claimed range and fleet size, compare measured latency/PDR against the model, and expand only while the documented margins remain positive.

Only after those steps may the new measurements replace the reference-profile numbers in a deployment document. Until then, the software may be generic, but the evidence is not.

4. Turn Assumptions into Executable Checks

All tools are in `sw/tools/mesh/`. Run them from the repository root. They use Python 3 and return nonzero when an acceptance gate fails.

4.1 `mesh_link_budget.py`: can one hop really close?

This is the deterministic RF calculator. It uses free-space loss where valid, two-ray propagation with a complex ground reflection coefficient, surface roughness, bank-angle polarisation loss, fade margin, Fresnel clearance, and radio horizon.

```
# Original area
python3 sw/tools/mesh/mesh_link_budget.py \
  --area-side 3000 --gcs-antenna-height 4

# Future square
python3 sw/tools/mesh/mesh_link_budget.py \
  --area-width 11000 --area-height 11000 --gcs-antenna-height 4

# Future strip
python3 sw/tools/mesh/mesh_link_budget.py \
  --area-width 17000 --area-height 3000 --gcs-antenna-height 4
```

Read **worst case margin** first. Positive margin means a direct corner-to-corner link closes after the modeled fade reserve; negative margin means multi-hop placement is required. A negative corner margin does not mean the mesh fails: it means no mission plan may leave a gap between neighboring routing aircraft larger than the applicable one-hop design range. For this profile that is at most 12.30 km air-to-air and 9.17 km to the 4 m GCS mast, before adding any mission-specific margin.

4.2 `mesh_phase_optimizer.py`: will the channel survive the fleet?

This tool parses the real PPRZLink XML, computes each message's configured PHY airtime, multiplies it by the reference-profile flood tax, solves Paparazzi phase offsets, replays the generated tick condition, simulates the configured radio queue, and writes the telemetry XML. A new radio requires new model inputs or model code before this result is meaningful.

The exact delivered `mesh` and `mesh_manifold` gates are:

```
python3 sw/tools/mesh/mesh_phase_optimizer.py \
--ac-ids 0,3,19,42,58,77,101,125,140,168,203,222,251 \
--relay-nodes 13 --nb-slots 32 --fair-slots 25 \
--mesh-period 2 --superframe 16 --max-reuse 8 \
--tdma-gate-frequency 100 --period-scale 1.5 --utilisation 0.60

python3 sw/tools/mesh/mesh_phase_optimizer.py \
--ac-ids 0,3,19,42,58,77,101,125,140,168,203,222,231,237,241,247,251 \
--relay-nodes 17 --nb-slots 32 --fair-slots 25 \
--mesh-period 2 --superframe 16 --max-reuse 8 \
--tdma-gate-frequency 100 --period-scale 3 --period FBW_STATUS=96 \
--utilisation 0.60
```

The AC_ID list is deliberately irregular. Its length drives population; the numeric values do not drive slot assignment. Do not hand-edit generated phase values. Change the model inputs and rerun the tool.

4.3 mesh_slot_sim.py: do joins, departures, and collisions settle?

This is a line-by-line Python mirror of the airborne slot allocator. It includes AC_ID 0, arbitrary IDs, 12 aircraft, dynamic state faults, emergency priority, cache pressure, and random land/rejoin churn.

```
for seed in 1 2 3 4 5 6 7 8; do
python3 sw/tools/mesh/mesh_slot_sim.py \
--frames 4000 --churn 0.05 --seed "$seed" --max-reuse 8 || exit 1
done
```

Acceptance means every node retains a slot, nobody exceeds the reuse cap, collision occupancy stays below 4%, every collision is bounded, and no deadlock survives a lease.

4.4 mesh_link_sim.py: does motion break the promise?

This packet-level Monte Carlo simulator imports the same RF model, moves the aircraft between waypoints, applies bank loss and correlated log-normal shadowing, and propagates each frame through an arbitrary number of routing hops. The reached set models the reference radio's duplicate filter.

```
# Nominal fleet and area
python3 sw/tools/mesh/mesh_link_sim.py \
--ac-ids 0,3,19,42,77,101,125,168,251 \
--width 3000 --height 3000 --duration 600 --seed 1

# Maximum fleet, future square
python3 sw/tools/mesh/mesh_link_sim.py \
--ac-ids 0,3,19,42,58,77,101,125,140,168,203,222,251 \
--width 11000 --height 11000 --duration 1200 --seed 7

# Maximum fleet, future strip
python3 sw/tools/mesh/mesh_link_sim.py \
--ac-ids 0,3,19,42,58,77,101,125,140,168,203,222,251 \
--width 17000 --height 3000 --duration 1200 --seed 7
```

The report shows direct and multi-hop PDR per aircraft, worst instantaneous margin, and busiest-radio duty. In one 17 x 3 km baseline seed, direct fleet PDR was 0.951 while modeled all-router multi-hop PDR was 1.000. Across the sixteen recorded expanded-area regression runs, all modeled relayed PDRs were 1.000; this is evidence that multi-hop helps

under those seeds and assumptions, not a guarantee for arbitrary geometry or terrain.

Simulation is not a proof that every random placement is connected. Before an expanded-area flight, run many seeds and enforce mission geometry that keeps a chain of aircraft inside one-hop range.

4.5 e52_provision.py: is the reference hardware actually identical?

Use `--dry-run` to inspect a profile without hardware, `--verify-only` to read back a reference radio, and `--factory-reset` before programming a known profile.

```
python3 sw/tools/mesh/e52_provision.py --ac-id 0 --dry-run
python3 sw/tools/mesh/e52_provision.py --ac-id 251 --dry-run
python3 sw/tools/mesh/e52_provision.py \
  --port /dev/ttyUSB0 --ac-id 251 --verify-only
```

Check that every reference radio reports a unique `SRC_ADDR=1000+AC_ID`, `TYPE=0`, rate 0, channel 24, 10 dBm conducted power for a 0 dBi antenna, broadcast option 3, and 460800 8N1.

4.6 No single green checkmark proves a network

No single tool validates the network. Each removes a different failure mode:



Passing a box means only that box's assumptions held. In particular, the link simulator assumes collision-free originations after `mesh_slot_sim.py` has validated the allocator; it does not reproduce radio firmware timing, adjacent channel interference, terrain blockage, UART faults, or antenna installation.



5. Inside the Flight Software

5.1 One compact state packet

The compact datalink message is 22 payload bytes and 30 PPRZLink wire bytes:

- flags: unified flight mode, rotorcraft, valid position, airborne, alert, emergency;
- clock mode: GPS, bounded holdover, or asynchronous fallback;
- latitude and longitude at 1e-7 degree;
- ellipsoid altitude in centimeters;
- packed course, ground speed, and climb rate.
- absolute recovery frame, zero outside `RECOVERY`.

The PPRZLink sender header is the source of truth for AC_ID. The payload does not duplicate it, so a relayed frame cannot claim a different aircraft without rewriting and rechecksumming the whole PPRZLink frame.

Message edits belong in `conf/messages_mesh_new.xml`. Do not edit

`sw/ext/pprzlink/message_definitions/v1.0/messages.xml`; it is external source, and the active `conf/messages.xml` resolves to the mesh definition.

5.2 Reuse the state estimator, do not invent another one

No parallel estimator or flight-mode model was invented. `traffic_info` uses:

- current `gps.fix` and `gps_tow_from_sys_ticks()` for GNSS time; configured fix-grace time is deliberately excluded from clock authority;
- an anchored monotonic clock for bounded holdover;
- `stateIsGlobalCoordinateValid()` and `stateGetPositionLla_i()` for the estimator's position in the initialized global frame;
- `stateGetHorizontalSpeedDir_f()`, `stateGetHorizontalSpeedNorm_f()`, and `stateGetSpeedEnu_f()` for motion;
- `autopilot_in_flight()`, `autopilot_get_mode()`, and `autopilot_throttle_killed()` for flight state;
- `electrical.vsupply` and `LOW_BAT_LEVEL` where available;
- Paparazzi `Min / Max`, geodetic conversion, and periodic telemetry APIs.

All mesh storage is static. There is no heap allocation, recursion, or variable-length array in the new path.

Clock validity and kinematic validity are separate. Losing GNSS does not immediately end TDMA because holdover preserves the epoch. For 16 seconds after the last observed 3D fix, `MESH_STATE` may continue carrying the estimator's globalized position and local velocity. This permits short inertial, airspeed, magnetometer, and barometer dead-reckoning gaps without freezing coordinates. After that configurable grace, the valid-position flag clears automatically; the packet still carries identity, flight state, and clock mode, while TCAS refuses the stale geometry. Increase `MESH_POSITION_HOLDOVER_MS` only from measured estimator error bounds, and never beyond the 60-second clock holdover. An estimator that has never acquired a shared global origin cannot emit valid mesh position. Vision, UWB, or SLAM may be used only when it supplies that shared frame and bounded uncertainty; unrelated local coordinates must never be encoded as LLA.

5.3 Back-pressure before the queue bites

`mesh_local_tx_in_flight()` is a leaky-bucket estimate of locally submitted frames which may not yet have drained. `traffic_info_mesh_periodic()` refuses a new `MESH_STATE` origination at depth three; state-aware reuse also contracts by one. Reference-radio internal relay frames and ordinary generated telemetry are not included in that estimate, and the reference command `AT+BACK=0` intentionally suppresses radio status text on the flight UART. This is therefore local admission control, not a physical cache-depth measurement. The separate optimizer acceptance ceiling of three applies to modeled total depth and reserves two of the five reference-hardware entries. The EByte manual states that an overflow forcibly clears every buffered frame. Another radio needs its own measured queue model and may require different bounds.

Raising either value does not add airtime or TDMA opportunities. In the dense 17-peer schedule, sweeping the optimizer acceptance ceiling from one through five leaves the same modeled peak depth of one, 59.4% channel utilization, and zero overflows. A higher airborne high-water mark can matter only during an unexpected local burst, precisely when

unobservable relay traffic makes the remaining margin valuable. Change it only after bench instrumentation shows total radio-queue occupancy and proves a repeatable delivery benefit under delayed UART service, clustered relays, commands, and topology churn.

5.4 Predict briefly, then fail closed

Each valid `MESH_STATE` stores its immutable position, velocity, local monotonic receive time, and mesh provenance in `traffic_info`. Safety consumers can ask for a caller-owned ENU snapshot projected with constant velocity:

Repeated calls always start from the original observation, so prediction never accumulates drift in the traffic table. Invalid-position heartbeats remain visible to TDMA membership but invalidate old kinematics immediately. Legacy traffic records keep their existing behavior and are never mesh-predicted.

TCAS bounds prediction and fresh decision-making to `TCAS_TAU_TA`, configured by the active airframe. After that interval an existing TA or RA and its altitude target are held, but no stale state opens, closes, or changes an advisory. After two complete superframes plus one slot and one 1 Hz task interval, currently 33.5 s, the track becomes `TCAS_UNAVAILABLE`; data loss is never reported as geometric resolution or `TCAS_NO_ALARM`. Clock arithmetic uses local monotonic age, while TDMA epochs use GPS week plus TOW. A synchronization change or a GPS-to-monotonic phase correction over 10 ms clears learned slot ownership and starts a new listen-before-claim cycle, including corrections that remain inside the same superframe or land in exactly the next one.

This policy bridges short update gaps without pretending that constant-velocity prediction is certain during maneuvers. It is not TCAS flight qualification. Measured packet latency, maneuver envelopes, loss bursts, and hardware-in-the-loop conflict scenarios remain required before operational use.

6. From Model to Flight: The Acceptance Ladder

Run these before flight after any message, telemetry, timing, routing, or radio change:

```

# 1. Channel, flood span, phases, and cache
python3 sw/tools/mesh/mesh_phase_optimizer.py \
  --ac-ids 0,3,19,42,58,77,101,125,140,168,203,222,251 \
  --relay-nodes 13 --nb-slots 32 --fair-slots 25 \
  --mesh-period 2 --superframe 16 --max-reuse 8 \
  --tdma-gate-frequency 100 --period-scale 1.5 --utilisation 0.60

python3 sw/tools/mesh/mesh_phase_optimizer.py \
  --ac-ids 0,3,19,42,58,77,101,125,140,168,203,222,231,237,241,247,251 \
  --relay-nodes 17 --nb-slots 32 --fair-slots 25 \
  --mesh-period 2 --superframe 16 --max-reuse 8 \
  --tdma-gate-frequency 100 --period-scale 3 --period FBW_STATUS=96 \
  --utilisation 0.60

# 2. Dynamic topology, fair quotas, clock steps, and state-aware contraction
for seed in 1 2 3 4 5 6 7 8; do
  python3 sw/tools/mesh/mesh_slot_sim.py \
    --frames 4000 --churn 0.05 --seed "$seed" --max-reuse 8 || exit 1
done

# 2a. Automatic mesh/mesh_manifold/mesh_solo mode policy
tests/utills/test_mesh_mode_policy.run

# 2b. Optimizer parser and airtime-accounting regressions
python3 -m unittest sw/tools/mesh/test_mesh_phase_optimizer.py

# 2c. Independent clocks, bounded holdover, and randomized fallback
python3 sw/tools/mesh/mesh_gps_denied_sim.py \
  --seeds 100 --workers 32 --duration 3600 --stress-ppm 100 --denied-nodes 13
python3 sw/tools/mesh/mesh_gps_denied_sim.py \
  --seeds 100 --workers 32 --duration 3600 --stress-ppm 100 --denied-nodes 6

# 3. RF geometry and moving multi-hop delivery
python3 sw/tools/mesh/mesh_link_budget.py \
  --area-width 17000 --area-height 3000 --gcs-antenna-height 4
python3 sw/tools/mesh/mesh_link_sim.py \
  --ac-ids 0,3,19,42,58,77,101,125,140,168,203,222,251 \
  --width 17000 --height 3000 --duration 1200 --seed 7

# 4. Embedded build
make CONF_XML=conf/userconf/OPENUAS/openuas_swarm_conf.xml \
  AIRCRAFT=Haydn ap.compile
make CONF_XML=conf/userconf/OPENUAS/openuas_swarm_conf.xml \
  AIRCRAFT=Adam ap.compile

# 5. Reference-radio profiles
python3 sw/tools/mesh/e52_provision.py --ac-id 0 --dry-run
python3 sw/tools/mesh/e52_provision.py --ac-id 251 --dry-run

# 6. Regenerate the paper figures (SVG and PNG)
python3 sw/tools/mesh/mesh_paper_plots.py

# 7. Build both styled, optimized PDF papers
python3 sw/tools/mesh/build_mesh_papers.py

```

The one-hour, 100-seed gate with 13 denied nodes and +/-100 ppm oscillator error produced 505720 originations, 6.915% collisions, no radio-queue flushes, and no unrecovered nodes. Denying six of 13 nodes produced 6.804% collisions, 48.358 s worst fallback convergence, no cache flushes, and no unrecovered nodes. Sparse-fleet runs averaged 5682, 2864, 1489, and 684 originations per node per hour for one, two, four, and nine nodes respectively. These are model results, not RF qualification; rerun the commands after changing any interval, flood span, clock, or recovery constant.

The slot simulator also checks 32-bit frame rollover, GPS-week rollover, same-frame and next-frame phase corrections, and GPS synchronization loss and reacquisition before each churn run.

Bench acceptance must also verify all thirteen radios together. Start them in different orders, remove up to six aircraft radios, restore them, and confirm:

- all present AC_IDs continue updating;
- the GCS (AC_ID 0) can transmit and route;
- no reference radio returns `OUT OF CACHE` ;
- removing any one routing node does not partition the remaining connected topology;
- radio readback matches the provisioning profile.

Record packet timestamps, source AC_ID, duplicate count where observable, end-to-end latency, and radio errors. A pass/fail result without these traces cannot distinguish RF loss, a slot collision, UART loss, cache flushing, or a topology partition.

Results and Evidence Summary

The table consolidates the quantitative claims and, critically, identifies the kind of evidence behind each one. Analytical and simulated outcomes are not reported as physical-radio measurements.

Question	Result	Evidence class	Interpretation
Does the nominal profile fit the operator ceiling?	57.7% modeled utilization for 13 peers	Analytical replay of generated telemetry	Passes the configured 60% ceiling under the reference flood model.
Does the dense profile fit the operator ceiling?	59.4% modeled utilization for 17 peers; modeled peak queue depth 1	Analytical optimizer and queue model	Passes narrowly; radio timing changes require re-solving.
Do leased slots recover under churn?	2.18-3.75% collision occupancy across the recorded 5% churn runs; all collisions bounded	Discrete-event slot simulation	Supports convergence of the allocator model, not RF collision behavior.
Does fallback recover under severe clock loss?	505720 originations, 6.915% collisions, no modeled queue flushes or unrecovered nodes for 13 denied nodes	100-seed, one-hour-per-seed simulation at +/-100 ppm	Supports bounded degraded operation under the modeled clocks and traffic.
Does partial denial recover?	6.804% collisions and 48.358 s worst convergence with 6 of 13 nodes denied	100-seed clock-loss simulation	Supports mixed synchronized/asynchronous recovery in the software model.
Can relaying improve expanded-area delivery?	Direct PDR 0.951 versus modeled relayed PDR 1.000 in one 17 x 3 km seed; relayed PDR 1.000 in 16 recorded expanded-area runs	Mobility and propagation simulation	Demonstrates benefit under sampled assumptions, not universal connectivity.
Does ground recovery work on the exercised hardware path?	Approximately 5 s PONG cadence, correct ALIVE MD5, registration after server restart, and convergence after link restart	Hardware-in-the-loop integration observation	Confirms the identity/health recovery path that was exercised.
Is the complete physical fleet qualified?	Not yet	Thirteen-radio bench and staged flight evidence pending	No claim of full-fleet RF qualification or TCAS certification is made.

No null-hypothesis significance test is appropriate for the deterministic acceptance gates. Stochastic tools instead expose seeds, duration, fleet size, and explicit thresholds so distributions and worst observed values can be reproduced. Future radio campaigns should report confidence intervals for PDR, latency, outage duration, and convergence time rather than only pass/fail totals.

7. What This Design Does Not Promise

- 1. The reference broadcast has no selective relay election.** All reference `TYPE=0` peers forward a new broadcast once. Its automatic RSSI route selection applies to unicast, not to this broadcast state stream. Another radio may select relays, but its timing and failure modes must be modeled.
- 2. Connectivity still depends on geometry.** All-router capability cannot bridge an empty 17 km gap. The mission planner must keep a connected chain of aircraft with adequate one-hop margin.
- 3. Dense-fleet update rate is deliberately lower.** At 16 aircraft plus the GCS, the 25 fair slots provide one or two `MESH_STATE` updates per peer per 16 s. TCAS fails closed when a track exceeds its prediction horizon; this profile does not claim uninterrupted 16-aircraft collision-avoidance coverage.
- 4. The GCS command tail is unslotted.** Commands and adaptive PINGs remain rare CSMA traffic. Airtime is reserved for control and immediate responses, but a half-duplex radio cannot preempt a frame already on air.
- 5. The RF model is terrain-agnostic.** The 10 dB fade reserve covers generic shadowing, not a ridge, building, or forest wall. Survey the real site.
- 6. All nodes must share protocol constants and message layout.** The clock and recovery fields make `MESH_STATE` 30 bytes on wire. Mixed firmware can decode shifted fields incorrectly, so rebuild and deploy the whole fleet atomically. A mixed 16-slot/32-slot fleet will also collide.
- 7. Asynchronous fallback is degraded operation.** It preserves low-rate discovery without false TDMA authority; it does not guarantee collision-free delivery. The 100-seed software gate must be followed by thirteen-radio HIL.
- 8. No RTC is required, but oscillator drift remains bounded.** Continuous power lets local monotonic time preserve the startup GNSS epoch for short holdover and drive randomized access indefinitely. It cannot preserve collision-safe TDMA phase indefinitely without oscillator calibration.
- 9. Dead reckoning is deliberately time-limited.** IMU, airspeed, magnetometer, and barometer inputs can bridge a short GNSS outage, but their unbounded drift cannot support indefinite TCAS geometry. Presence and command transport continue after the valid-position flag expires.
- 10. Sixty-four aircraft is graceful-degradation research scope.** AC_IDs and software membership support the range, but the present 32-slot all-router channel cannot provide collision-free 64-aircraft state or real-time TCAS. Do not advertise 64-aircraft flight safety from this profile.

The design preference is explicit: when speed and redundancy conflict, keep the all-routing topology and reduce originated telemetry first. The optimizer is the authority for that trade, and it must exit zero before flight.

Threats to Validity

Construct validity. Channel utilization, queue depth, collision occupancy, PDR, and convergence time are proxies for network usefulness. They do not by themselves establish safe separation, command availability, or acceptable operator workload. TCAS certification is explicitly outside the claim.

Internal validity. Several tools share constants and assumptions with the implementation. Agreement can therefore reproduce the same modeling error. Independent packet traces, radio queue instrumentation, and a channel emulator are needed to break that dependency. The propagation model omits terrain, installation-specific antenna patterns, external interference, and unmodeled radio firmware behavior.

External validity. Quantitative results apply to the stated fleet sizes, traffic profiles, geometry, oscillator bounds, and E52 reference measurements. They do not transfer automatically to another radio, regulatory region, airframe, antenna installation, or mission. Section 3 defines the required requalification.

Reliability and repeatability. Seeded simulations and generated telemetry make software results repeatable, but the recorded hardware observations are not yet a statistically powered fleet experiment. The pending thirteen-radio bench must publish raw traces, environmental conditions, firmware revisions, and repeated-run dispersion.

8. Flying It Without Becoming the Network Scheduler

No radio reprovisioning between scenarios, additional application, or manual telemetry-mode change is needed. Build and flash aircraft with a telemetry profile containing `mesh`, `mesh_manifold`, and `mesh_solo`, then start the normal link/server/GCS session. Every peer keeps the same qualified broadcast profile. For the tested EByte deployment, that means all-routing broadcast at 62.5 kbit/s with a 460800-baud UART.

The aircraft starts in `mesh` and changes to `mesh_solo` after GCS contact and the 16-second peer-quiet interval. Starting another aircraft immediately leaves solo mode; joining and departing aircraft move the fleet between `mesh` and `mesh_manifold` with 12-peer/10-peer hysteresis. All transitions are automatic and invisible to the operator.

The supplied fixed-wing profiles use the generated `Ap` process. The dedicated `openuas_rotorcraft_mesh.xml` profile uses `Main` and native `ROTORCRAFT_FP`, so the selector itself is firmware-neutral. The delivered mesh profiles define all three modes. Downstream two-mode profiles remain compatible and continue using `mesh` telemetry where a specialized mode is absent.

Validation includes fixed-wing and rotorcraft builds, pure mode-policy tests, mesh churn and GPS-loss simulations, single-aircraft rate measurements, and a two-aircraft fallback run through the standard OCaml link. The final identity and health protocol was also exercised on hardware: PONG remained approximately five seconds apart, ALIVE carried the expected MD5, server restart recovered registration, and link restart converged from the server's `AIRCRAFTS` snapshot.

9. Conclusion: Closing the Loop

The abstract began with a practical problem: a fleet of small aircraft must share fresh state and occasional commands over one constrained channel while members join, leave, lose GNSS, or restart. The implementation addresses each part of that problem with a bounded mechanism rather than an assumption:

- learned, leased slots replace fixed aircraft-ID schedules;
- fair-share quotas, phase solving, and radio-queue back-pressure bound channel and buffer demand as the fleet changes;
- GNSS TDMA, monotonic holdover, fleet-wide asynchronous fallback, and staged recovery prevent an invalid clock from masquerading as synchronization;
- compact state snapshots and fail-closed freshness rules prevent old geometry from being interpreted as current collision-avoidance evidence;
- separate health and identity exchanges, durable registration snapshots, and restart-safe serial handling make the ground system converge after loss;
- executable RF, airtime, queue, churn, and clock-loss models turn radio assumptions into repeatable acceptance checks.

The resulting system therefore satisfies the abstract's engineering objective within the documented 13/17-peer reference envelope: it provides automatic, fault-aware broadcast state exchange without static slot maps or an operator acting as network coordinator. Hardware tests and representative builds support that conclusion. The claim remains deliberately bounded. It is not universal radio compatibility, unrestricted fleet scaling, or TCAS flight certification. Those require the requalification ladder in Sections 3 and 6. In other words, the problem is fixed by making uncertainty explicit and testable, not by declaring that wireless networks have finally agreed to behave.

10. Where the Work Goes Next

These are the three highest-value next steps. They are deliberately not described as current capabilities.

10.1 Put the complete fleet on the bench

The software models now agree, but the largest remaining uncertainty is radio firmware itself: forwarding jitter, duplicate-filter behavior, cache flushes, UART buffering, CSMA interaction, and recovery after a radio resets. Build a repeatable thirteen-radio test fixture before increasing area or traffic rate.

The fixture should use conducted RF paths, attenuators or a channel emulator where possible, rather than thirteen nearby antennas at full signal. Exercise:

- simultaneous and staggered power-up;
- one through six node removals and rejoins;
- asymmetric links and a forced three- or four-hop chain;
- fading near receiver sensitivity;
- emergency-state bursts, GCS commands, and sustained maximum telemetry;
- UART interruption, radio reset, and duplicate-filter expiry.

Acceptance should bound per-node PDR, 95th and 99th percentile latency, longest outage, convergence time after churn, cache high-water events, and recovery without manual reprovisioning. Use the measured forwarding-delay distribution to rerun `mesh_phase_optimizer.py`; replace the current statistical slot assumption only after the hardware data supports it.

10.2 Broadcast safety, unicast everything else

Keep compact safety state on broadcast because every peer needs it and it must survive without a coordinator. Move traffic with one intended recipient, such as commands, parameter exchange, logs, or bulk telemetry, toward a candidate radio's qualified unicast path so it can avoid paying the 13-way broadcast flood tax.

This is an architectural experiment, not a provisioning-only change. Determine first whether the chosen radio can switch destination and service safely at runtime at the required rate. If mode changes disrupt routing state or duplicate filters, use time-separated traffic classes or a second radio rather than rapidly rewriting radio configuration. Add sequence numbers, delivery policy, and bounded retries at the application layer; never allow bulk unicast to starve `MESH_STATE` or emergency traffic. Re-run airtime, cache, churn, and hardware tests for the mixed profile.

10.3 Know when one aircraft holds the network together

All-router flooding provides a forwarding opportunity, but connectivity is a property of the instantaneous RF graph. In a long strip, one aircraft can be an articulation node: removing it divides the graph even though every remaining radio is configured as a router.

A future topology service should estimate link quality, construct the live connectivity graph, identify articulation nodes and weak edges, and expose a margin-to-partition warning to the GCS and mission planner. Position-derived range is a useful first estimate, but measured RSSI/PDR history is preferable because antenna shadowing and interference are not visible in geometry alone.

The response must match what the current radio can actually control. In the reference broadcast design Paparazzi cannot elect which radio forwards a packet. It can:

- warn or constrain mission geometry before a critical bridge disappears;
- protect the state-update opportunity of aircraft that maintain connectivity;
- request repositioning or loiter points that create a second path;
- feed route quality into the future unicast traffic class.

The target is at least two node-disjoint GCS paths for safety-critical mission regions where geometry permits. When that is impossible, the system should report the single-node failure explicitly instead of treating “all nodes route” as proof of redundancy.

Reproducibility and Data Availability

The implementation, configuration, protocol definitions, simulators, tests, and exact acceptance commands are contained in this repository. Sections 4 and 6 specify executable entry points and parameters. Stochastic tools accept explicit seeds; generated telemetry must be regenerated from model inputs rather than edited manually. The tested reference-radio manual is archived in this documentation directory so provisioning claims can be checked against the same revision.

The numerical results reported here are generated summaries. Raw multi-radio packet traces and a complete environmental dataset are not yet available because the thirteen-radio campaign remains future work. Any archival release should include source revision, tool and Python versions, command lines, seeds, generated configurations, radio firmware and readback, antenna and power setup, geometry, weather or channel-emulator settings, and timestamped packet-level data. These are minimum provenance requirements for independent replication.

Declarations

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Ethics and safety: The reported work concerns software, simulation, and engineering hardware tests and includes no human or animal subjects. Flight use remains subject to applicable aviation, spectrum, institutional, and operational safety approvals. Nothing in this paper constitutes certification of the mesh or TCAS behavior.

11. Literature and Engineering Sources

This is an annotated reading list, not a claim that the implementation conforms to every protocol cited. Each source either supplies a design precedent, defines an interface used by the implementation, or records evidence for the tested reference profile.

1. International Telecommunication Union, [*Recommendation ITU-R M.1371: Technical characteristics for an automatic identification system using time division multiple access in the VHF maritime mobile frequency band*](#). Its self-organising TDMA mechanisms provide the principal precedent for observing slots, announcing use, and expiring leases without a central scheduler. This paper adapts the ideas; it does not implement AIS.
2. P. Levis, T. Clausen, J. Hui, O. Gnawali, and J. Ko, [*The Trickle Algorithm*](#), RFC 6206, March 2011. Trickle is the standard reference for suppressing redundant transmissions while reacting quickly to inconsistency, a useful comparison for membership and future dissemination work.

3. T. Clausen and P. Jacquet, [Optimized Link State Routing Protocol \(OLSR\), RFC 3626](#), October 2003. OLSR formalizes duplicate-aware flooding, multipoint relays, link hysteresis, and topology maintenance in mobile ad hoc networks; it frames the selective-relay alternative to the reference radio's all-router broadcast.
4. T. Winter et al., [RPL: IPv6 Routing Protocol for Low-Power and Lossy Networks, RFC 6550](#), March 2012. RPL is relevant for its treatment of constrained, unstable links, freshness, repair, monitoring, and explicit separation of protocol goals from link-layer behavior.
5. Paparazzi UAV contributors, [Paparazzi UAS documentation](#). Paparazzi supplies the flight architecture, generated messages, telemetry, simulation, and ground segment into which this mesh is integrated; repository source remains authoritative for the exact version described here.
6. EByte, [E52-xxxNWxxS User Manual, version 1.4](#). This local manual is the evidence source for commands, queue behavior, power, and forwarding features of the E52-400NW22S tested reference hardware. Its product-specific properties are not requirements of the generic architecture.
7. M. Haenggi, *Stochastic Geometry for Wireless Networks*, Cambridge University Press, 2012. The text provides broader foundations for reasoning about spatial wireless connectivity and interference beyond a single deterministic link budget.
8. D. B. West, *Introduction to Graph Theory*, 2nd ed., Prentice Hall, 2001. Articulation vertices and node-disjoint paths give the precise language used in Section 10.3 to distinguish configured forwarding from actual topological redundancy.
9. Mark Twain, [Eve's Diary](#), 1906. The acknowledgements quote its argument for experiment over conjecture. Here that principle is operational: simulations, builds, traces, and radio readback are acceptance evidence, while untested radio substitutions are not.
10. I. F. Akyildiz, X. Wang, and W. Wang, "Wireless mesh networks: a survey," *Computer Networks*, vol. 47, no. 4, pp. 445-487, 2005. [doi:10.1016/j.comnet.2004.12.001](#). This foundational survey covers mesh architectures, protocol layers, self-healing topologies, and the research challenges against which this deliberately narrower airborne broadcast design can be compared.
11. A. Raniwala and T. C. Chiueh, "Architecture and algorithms for an IEEE 802.11-based multi-channel wireless mesh network," in *Proceedings of IEEE INFOCOM 2005*, vol. 3, pp. 2223-2234, 2005. [doi:10.1109/INFCOM.2005.1498511](#). The Hyacinth architecture demonstrates distributed channel assignment and load-balanced routing for multi-interface nodes. It is a future multi-channel comparison, not a description of the present single-radio profile.
12. D. S. De Couto, D. Aguayo, J. Bicket, and R. Morris, "A high-throughput path metric for multi-hop wireless routing," in *Proceedings of the 9th Annual International Conference on Mobile Computing and Networking (MobiCom '03)*, pp. 134-146, 2003. [doi:10.1145/938985.939000](#). Expected Transmission Count (ETX) replaces hop count with measured delivery probability. It supports the proposed use of PDR history when estimating weak links and live topology.
13. K. Uemura, L. Barolli, and M. Takizawa, "A Delaunay edges and simulated annealing-based integrated approach for mesh router placement optimization in wireless mesh networks," *Sensors*, vol. 23, no. 3, article 1050, 2023. [doi:10.3390/s23031050](#). Its geometric and stochastic optimization of router placement is relevant to mission planning for connected formations, although moving aircraft add dynamics absent from static router placement.
14. D. Benyamina, A. S. Hafid, and M. Gendreau, "Wireless mesh networks design—A survey," *IEEE Communications Surveys & Tutorials*, vol. 14, no. 2, pp. 299-310, 2012. [doi:10.1109/SURV.2011.031811.00073](#). This survey organizes gateway placement, backbone topology, and cost-versus- quality optimization, providing context for the geometry and redundancy requirements in Section 10.3.
15. J. Tang, A. Sen, and X. Zhang, "Cross-layer optimization for throughput maximization in multi-channel wireless mesh networks," *IEEE Transactions on Wireless Communications*, vol. 5, no. 6, pp. 1506-1516, 2006. [doi:10.1109/TWC.2006.1638666](#). Its joint channel-assignment, scheduling, and routing formulation illustrates why these concerns cannot be optimized independently when a future platform adds multiple channels.

16. P. Kyasanur and N. H. Vaidya, "Capacity of multi-channel wireless networks: Impact of number of channels and interfaces," *IEEE Transactions on Mobile Computing*, vol. 5, no. 5, pp. 471-487, 2006. [doi:10.1109/TMC.2006.1613854](https://doi.org/10.1109/TMC.2006.1613854). The capacity bounds clarify that additional channels help only in relation to the number of usable interfaces. They prevent assuming that spectrum alone removes a single-radio switching bottleneck.
17. P. Gupta and P. R. Kumar, "The capacity of wireless networks," *IEEE Transactions on Information Theory*, vol. 46, no. 2, pp. 388-404, 2000. [doi:10.1109/18.825799](https://doi.org/10.1109/18.825799). This landmark analysis establishes fundamental multi-hop scaling limits. Its central lesson applies directly here: fleet growth consumes per-node capacity even when routing and scheduling are well designed.
18. A. Salama, A. Stergioulis, S. A. Zaidi, and D. McLernon, "Decentralized federated learning on the edge over wireless mesh networks," *arXiv preprint arXiv:2311.01186*, (2023). [doi:10.48550/arXiv.2311.01186](https://doi.org/10.48550/arXiv.2311.01186). This work studies decentralized learning over interference-limited multi-hop meshes. It is relevant to possible distributed adaptation, but learned control is outside the current deterministic flight implementation.
19. L. H. Binh and T. V. T. Duong, "A novel and effective method for solving the router nodes placement in wireless mesh networks using reinforcement learning," *PLoS ONE*, vol. 19, no. 4, article e0301073, 2024. [doi:10.1371/journal.pone.0301073](https://doi.org/10.1371/journal.pone.0301073). Its reinforcement-learning treatment of router placement offers a possible future approach to formation geometry, subject to hard safety constraints and validation outside the learning loop.
20. R. Draves, J. Padhye, and B. Zill, "Routing in multi-radio, multi-hop wireless mesh networks," in *Proceedings of the 10th Annual International Conference on Mobile Computing and Networking (MobiCom '04)*, pp. 114-128, (2004). [doi:10.1145/1023720.1023732](https://doi.org/10.1145/1023720.1023732). Weighted Cumulative Expected Transmission Time (WCETT) combines link quality, rate, and intra-flow interference. It is a useful metric precedent if the future unicast path gains multiple radios or channels.
21. P. Bahl, R. Chandra, and J. Dunagan, "SSCH: Slotted seeded channel hopping for capacity improvement in wireless LANs," in *Proceedings of the 10th Annual International Conference on Mobile Computing and Networking (MobiCom '04)*, pp. 216-230, 2004. [doi:10.1145/1023720.1023742](https://doi.org/10.1145/1023720.1023742). SSCH shows how a single radio can exploit non-overlapping channels through coordinated hopping. Such a design would require a new synchronization, rendezvous, and failure analysis before use in this mesh.
22. Y. Watanabe, A. Fujiwara, and N. Kato, "A novel routing control method using federated learning in large-scale wireless mesh networks," *IEEE Transactions on Wireless Communications*, vol. 22, no. 12, pp. 9291-9300, (2023). [doi:10.1109/TWC.2023.3269785](https://doi.org/10.1109/TWC.2023.3269785). The paper applies federated learning to distributed routing and congestion control. It belongs to the longer-term adaptive-routing literature rather than the evidence base for the present bounded algorithms.
23. P. H. Pathak and R. Dutta, "A survey of network design problems and joint design approaches in wireless mesh networks," *IEEE Communications Surveys & Tutorials*, vol. 13, no. 3, pp. 396-428, 2011. [doi:10.1109/SURV.2011.060710.00062](https://doi.org/10.1109/SURV.2011.060710.00062). Its review of placement, power control, scheduling, and routing supports the paper's qualification philosophy: changing one layer requires re-evaluating the coupled airtime, connectivity, queue, and safety constraints.